

# GCSE Mathematics

Predicted Paper 2 — Higher Tier

Set B

Pearson Edexcel style (1MA1 / 2H)

## Calculator

Time: 1 hour 30 minutes · Total marks: 80

## Materials required

- Calculator
- Ruler graduated in centimetres and millimetres
- Protractor
- Pair of compasses
- Pen, HB pencil, eraser

## Instructions

- Answer **all** 22 questions in the spaces provided.
  - You must show all your working.
- Diagrams are NOT accurately drawn unless stated otherwise.
  - Calculators may be used.
- If your calculator does not have a  $\pi$  button, take  $\pi$  to be 3.142.

## Advice

- Read each question carefully before you start.
- The marks for each question are shown in brackets — use them as a time guide.
- Try to answer every question. Check your answers if you have time at the end.

*This is an original predicted paper produced by GCSEMathsAI for revision purposes only. It is not affiliated with or endorsed by Pearson Edexcel.*

**Answer ALL questions. Write your answers in the spaces provided.**

- 1.** Tom buys a watch for £180. He sells it for £234. Work out the percentage profit Tom makes.

*(3 marks)*

- 2.** Work out  $(5 \times 10^4) + (3 \times 10^3)$ . Give your answer in standard form.

*(2 marks)*

- 3.** A 450 g jar of jam costs £2.85. A 720 g jar of the same jam costs £4.32. Which jar is the better value for money? Show your working clearly.

*(3 marks)*

- 4.** Solve the simultaneous equations:

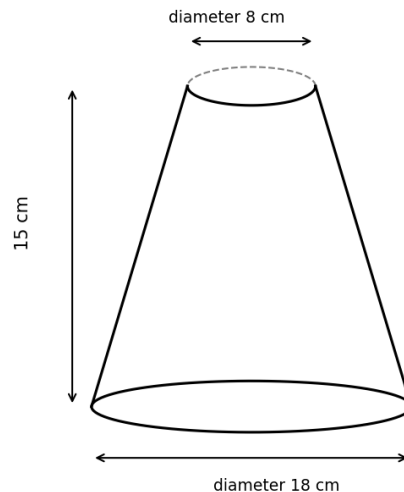
$$3x + 2y = 17$$

$$5x - 2y = 7$$

*(3 marks)*

5. The diagram shows a frustum formed by removing a smaller cone from the top of a larger cone. The larger cone has a base diameter of 18 cm. The smaller cone (the top) has a diameter of 8 cm. The total height of the frustum is 15 cm. Work out the volume of the frustum. Give your answer correct to 3 significant figures.

(5 marks)



6. Solve  $x^2 + 5x - 2 = 0$ . Give your solutions correct to 3 significant figures. Show your working clearly.

(3 marks)

7. There are 60 students in Year 11. Of these students: 32 study French, 24 study German, 10 study both French and German.

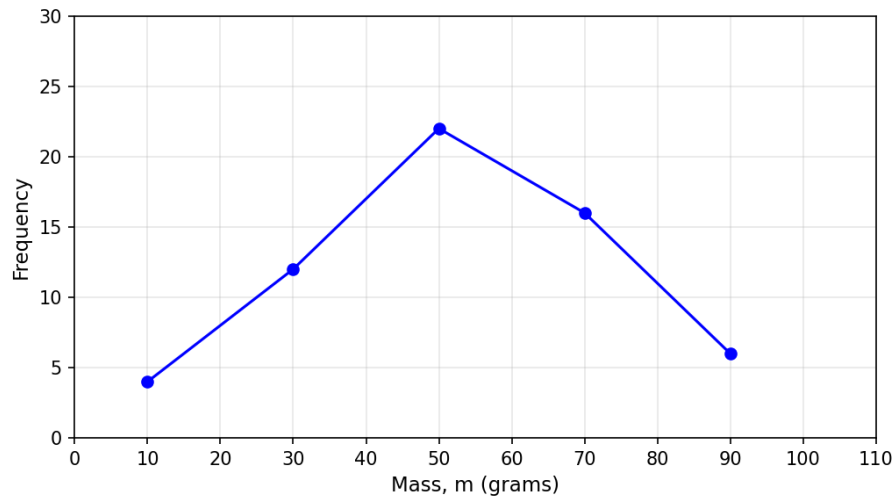
(a) Work out the number of students who study neither French nor German.

(2 marks)

(b) One of the 60 students is chosen at random. Given that the chosen student studies French, work out the probability that the student also studies German.

(2 marks)

8. The frequency polygon shows information about the masses, in grams, of 60 apples.



(a) Use the frequency polygon to find the modal class.

(1 marks)

(b) Work out an estimate for the mean mass of an apple.

(4 marks)

9. Simplify fully:

(a)  $(3a^2b^4)^3$

(2 marks)

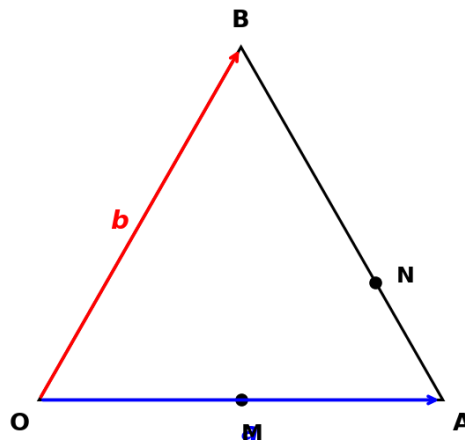
(b)  $(16x^8)^{3/4}$

(2 marks)

10. Rationalise the denominator and simplify:  $6 / (3 - \sqrt{3})$ . Give your answer in the form  $a + b\sqrt{3}$  where  $a$  and  $b$  are integers.

(3 marks)

11. OAB is a triangle.  $\mathbf{OA} = \mathbf{a}$  and  $\mathbf{OB} = \mathbf{b}$ . M is the midpoint of OA. N is the point on AB such that  $AN : NB = 1 : 2$ .



(a) Express the vector  $\mathbf{MN}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$ . Simplify your answer.

(4 marks)

(b) Hence show that  $\mathbf{MN}$  is parallel to a vector of the form  $(1/6)(p\mathbf{a} + q\mathbf{b})$  where  $p$  and  $q$  are integers to be found.

(2 marks)



**12.** A bank account pays compound interest at a rate of 3.5% per year. After 4 years, the balance in the account is £1146.25. Work out the amount that was originally invested. Give your answer correct to the nearest pound.

(3 marks)

**13.** A radioactive substance decays exponentially. The mass  $M$ , in grams, after  $t$  years is given by the formula  $M = 80 \times (0.92)^t$ .

**(a)** Find the mass of the substance after 5 years. Give your answer to 1 decimal place.

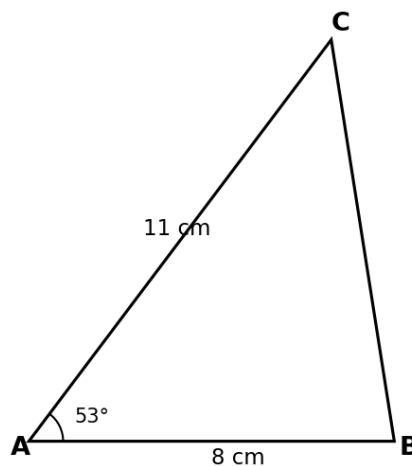
(2 marks)

**(b)** By how many percent does the mass decrease each year?

(1 marks)

**14.** The diagram shows triangle ABC.  $AB = 8$  cm,  $AC = 11$  cm and angle  $BAC = 53^\circ$ . Work out the length of BC. Give your answer correct to 3 significant figures.

(3 marks)



**15.** Write as a single fraction in its simplest form:  $(2x + 1) / (x - 2) - (x - 3) / (x + 1)$ .

(4 marks)

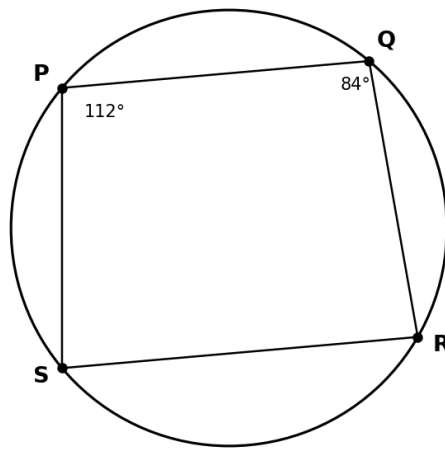




**16.** Show that the recurring decimal  $0.454545\dots$  (with the digits 45 repeating) can be written as the fraction  $\frac{5}{11}$ .

(3 marks)

**17.** PQRS is a cyclic quadrilateral (all four vertices lie on a circle). Angle  $SPQ = 112^\circ$  and angle  $PQR = 84^\circ$ .



**(a)** Work out the size of angle QRS. Give a reason for your answer.

(2 marks)

**(b)** Work out the size of angle RSP. Give a reason for your answer.

(2 marks)

**18.**  $y$  is directly proportional to the square root of  $x$ . When  $x = 36$ ,  $y = 24$ .

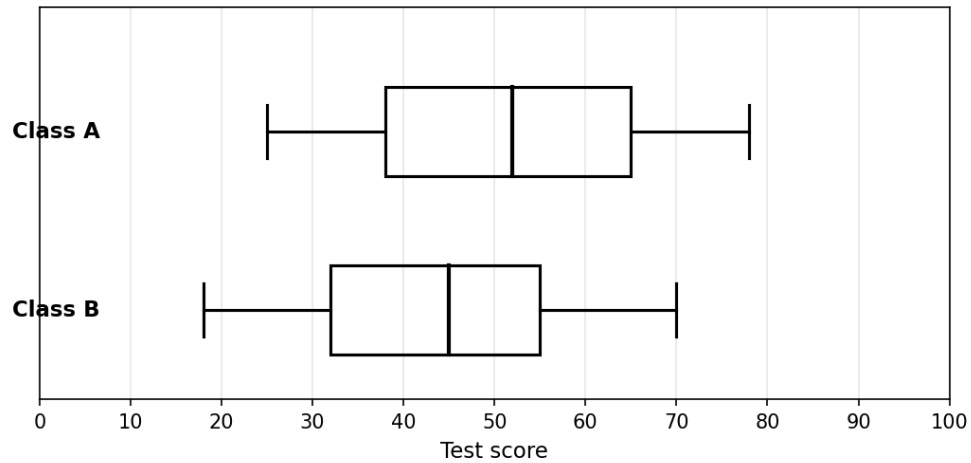
**(a)** Express  $y$  in terms of  $x$ .

(2 marks)

**(b)** Find the value of  $y$  when  $x = 81$ .

(2 marks)

**19.** The box plots show information about the test scores of two classes, Class A and Class B.



**(a)** Write down the median score for Class A.

*(1 marks)*

**(b)** Work out the interquartile range for Class B.

*(2 marks)*

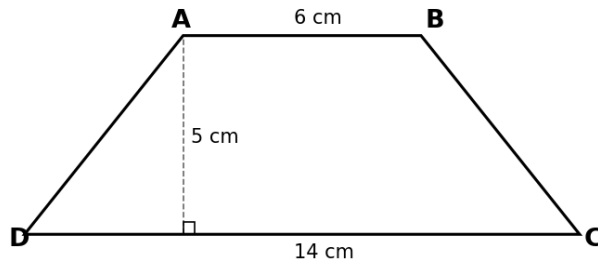
**(c)** Compare the test scores of the two classes. Make two clear statements.

*(2 marks)*

**20.** Solve the inequality  $x^2 - 5x - 14 > 0$ .

*(4 marks)*

**21.** The diagram shows trapezium ABCD. AB is parallel to DC.  $AB = 6\text{ cm}$ ,  $DC = 14\text{ cm}$  and the perpendicular distance between AB and DC is  $5\text{ cm}$ . AD is perpendicular to DC.



**(a)** Work out the area of trapezium ABCD.

*(2 marks)*

**(b)** Work out the length of BC. Give your answer correct to 3 significant figures.

*(3 marks)*

**22.** Prove algebraically that the product of any two consecutive even integers is always a multiple of 4.

*(4 marks)*

**TOTAL FOR PAPER: 80 MARKS**

**END**

# Mark Scheme — Predicted Paper 2 Set B

Higher Tier · Edexcel 1MA1 style · 80 marks

| Q     | Answer                                    | Mark scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | 30%                                       | M1: profit = 234 – 180 = 54. M1: $54/180 \times 100$ . A1: 30. <b>[3]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2     | $5.3 \times 10^4$                         | M1: 50000 + 3000 = 53000. A1: $5.3 \times 10^4$ . <b>[2]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3     | 720 g jar is better value                 | M1: 285/450 = 0.633p/g OR 432/720 = 0.6p/g (or g per £). M1: compares both. A1: 720 g jar is cheaper per gram. <b>[3]</b>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4     | x = 3, y = 4                              | M1: adds equations to eliminate y: 8x = 24. M1: x = 3, substitutes. A1: y = 4 (both correct). <b>[3]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5     | 2090 cm <sup>3</sup> (3 s.f.)             | M1: uses similar triangles to find the full cone height. Small cone radius : large cone radius = 4 : 9, so heights are in the same ratio. If full cone height = H and small cone height = h, then H – h = 15 and h/H = 4/9, giving H = 27 cm and h = 12 cm. M1: volume of full cone = $(1/3)\pi(9)^2(27) = 729\pi$ cm <sup>3</sup> . M1: volume of small cone = $(1/3)\pi(4)^2(12) = 64\pi$ cm <sup>3</sup> . M1: frustum volume = $729\pi - 64\pi = 665\pi$ . A1: $665\pi = 2089.5... \Rightarrow 2090$ cm <sup>3</sup> (3 s.f.). <b>[5]</b> |
| 6     | x = 0.372 or x = –5.37                    | M1: a = 1, b = 5, c = –2. M1: applies quadratic formula. A1: both solutions correct to 3 s.f. <b>[3]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7(a)  | 14 students                               | M1: students studying F or G = 32 + 24 – 10 = 46. A1: 60 – 46 = 14. <b>[2]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 7(b)  | 10/32 = 5/16                              | M1: identifies conditional probability = (both)/(French). A1: 10/32 = 5/16. <b>[2]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 8(a)  | 40 < m ≤ 60 (the 50 g class)              | B1: identifies the modal class as 40–60 g (using midpoint 50). <b>[1]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 8(b)  | ≈ 50.7 g                                  | M1: identifies midpoints 10, 30, 50, 70, 90. M1: $\Sigma fx = 40 + 360 + 1100 + 1120 + 540 = 3160$ . M1: $\Sigma f = 4 + 12 + 22 + 16 + 6 = 60$ . A1: $3160/60 \approx 52.7$ g. (Accept 52–53.) <b>[4]</b>                                                                                                                                                                                                                                                                                                                                    |
| 9(a)  | $27a^6b^{12}$                             | M1: cubes 3 (= 27) and indices ( $a^6$ , $b^{12}$ ). A1: all correct. <b>[2]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 9(b)  | $8x^6$                                    | M1: $16^{3/4} = 8$ AND $x^{8 \cdot 3/4} = x^6$ . A1: $8x^6$ . <b>[2]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 10    | $3 + \sqrt{3}$                            | M1: multiplies top and bottom by $(3 + \sqrt{3})$ . M1: denominator = $9 - 3 = 6$ ; numerator = $6(3 + \sqrt{3}) = 18 + 6\sqrt{3}$ . A1: $3 + \sqrt{3}$ , so a = 3, b = 1. <b>[3]</b>                                                                                                                                                                                                                                                                                                                                                         |
| 11(a) | $MN = -(1/3)a + (1/3)b$                   | M1: $OM = (1/2)a$ . M1: $ON = a + (1/3)(b - a) = (2/3)a + (1/3)b$ . M1: $MN = ON - OM$ . A1: $(1/6)(b - a)$ or equivalent (e.g., $-(1/3)a + (1/3)b$ ). Note: $(1/6)(-2a + 2b) = (1/3)(b - a)$ . Either form accepted. <b>[4]</b>                                                                                                                                                                                                                                                                                                              |
| 11(b) | $MN = (1/6)(-2a + 2b)$ , so p = –2, q = 2 | M1: rewrites in required form. A1: identifies p = –2, q = 2 (or equivalent). <b>[2]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 12    | £1000                                     | M1: original $\times 1.035^4 = 1146.25$ . M1: original = $1146.25 / 1.035^4$ . A1: 1000.00 (to nearest pound). <b>[3]</b>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 13(a) | 52.6 g                                    | M1: $80 \times (0.92)^5$ . A1: 52.6 (1 d.p.). <b>[2]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 13(b) | 8%                                        | B1: $1 - 0.92 = 0.08 = 8\%$ . <b>[1]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 14    | BC = 8.84 cm (3 s.f.)                     | M1: cosine rule: $BC^2 = 8^2 + 11^2 - 2 \times 8 \times 11 \times \cos 53$ . M1: = 64 + 121 – 105.95... = 79.05. A1: $BC = \sqrt{79.05} \approx 8.89$ cm. (Accept 8.84–8.90 depending on precision used.) <b>[3]</b>                                                                                                                                                                                                                                                                                                                          |
| 15    | $(x^2 + 9x + 5) / ((x - 2)(x + 1))$       | M1: common denominator $(x - 2)(x + 1)$ . M1: numerator = $(2x + 1)(x + 1) - (x - 3)(x - 2)$ . M1: expands: $(2x^2 + 3x + 1) - (x^2 - 5x + 6)$ . A1: $x^2 + 8x - 5$ over $(x - 2)(x + 1)$ . Note: precise expansion gives $x^2 + 8x - 5$ in numerator. <b>[4]</b>                                                                                                                                                                                                                                                                             |

| Q     | Answer                                | Mark scheme                                                                                                                                                                                                                                                 |
|-------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16    | 5/11                                  | M1: lets $x = 0.454545\dots$ , so $100x = 45.4545\dots$ . M1: $100x - x = 45 \Rightarrow 99x = 45$ . A1: $x = 45/99 = 5/11$ . <b>[3]</b>                                                                                                                    |
| 17(a) | Angle QRS = $68^\circ$                | B1: $68^\circ$ . B1: reason "opposite angles of cyclic quadrilateral sum to $180^\circ$ " ( $180 - 112 = 68$ ). <b>[2]</b>                                                                                                                                  |
| 17(b) | Angle RSP = $96^\circ$                | B1: $96^\circ$ . B1: reason "opposite angles of cyclic quadrilateral sum to $180^\circ$ " ( $180 - 84 = 96$ ). <b>[2]</b>                                                                                                                                   |
| 18(a) | $y = 4\sqrt{x}$                       | M1: $y = k\sqrt{x}$ , $24 = k \times 6$ . A1: $k = 4$ , $y = 4\sqrt{x}$ . <b>[2]</b>                                                                                                                                                                        |
| 18(b) | $y = 36$                              | M1: $y = 4\sqrt{81} = 4 \times 9$ . A1: $y = 36$ . <b>[2]</b>                                                                                                                                                                                               |
| 19(a) | 52                                    | B1: median for Class A = 52. <b>[1]</b>                                                                                                                                                                                                                     |
| 19(b) | IQR = 23                              | M1: $Q3 - Q1 = 55 - 32$ . A1: IQR = 23. <b>[2]</b>                                                                                                                                                                                                          |
| 19(c) | Two valid comparisons                 | B1: comparison of medians/centre (e.g. Class A median higher, so on average Class A did better). B1: comparison of spread/IQR (e.g. Class A more spread out / wider IQR). <b>[2]</b>                                                                        |
| 20    | $x < -2$ or $x > 7$                   | M1: factorises $(x - 7)(x + 2)$ . M1: identifies critical values $-2$ and $7$ . M1: considers shape of parabola (positive coefficient, U-shape). A1: $x < -2$ or $x > 7$ . <b>[4]</b>                                                                       |
| 21(a) | $50 \text{ cm}^2$                     | M1: $(1/2)(6 + 14) \times 5$ . A1: 50. <b>[2]</b>                                                                                                                                                                                                           |
| 21(b) | $BC \approx 9.43 \text{ cm}$ (3 s.f.) | M1: identifies horizontal distance = $14 - 6 - 0 = 8 \text{ cm}$ . (Or recognises right triangle with legs 8 and 5.) M1: applies Pythagoras: $BC^2 = 8^2 + 5^2 = 89$ . A1: $BC = \sqrt{89} = 9.43 \text{ cm}$ . <b>[3]</b>                                  |
| 22    | Proof shown                           | M1: lets two consecutive even integers be $2n$ and $2n + 2$ . M1: product = $2n \times (2n + 2) = 4n^2 + 4n$ . M1: factorises as $4n(n + 1)$ or $4(n^2 + n)$ . A1: clearly identifies this as $4 \times \text{integer}$ , hence a multiple of 4. <b>[4]</b> |

# Worked Solutions — Predicted Paper 2 Set B

Higher Tier · Edexcel 1MA1 style

## Q1: Percentage profit

Profit = selling price – cost price =  $234 - 180 = £54$ .

Percentage profit = (profit / cost price)  $\times 100 = (54 / 180) \times 100 = \mathbf{30\%}$ .

## Q2: Standard form addition

Convert to ordinary numbers:  $5 \times 10^4 = 50\,000$ ;  $3 \times 10^3 = 3\,000$ .

Sum =  $53\,000 = \mathbf{5.3 \times 10^4}$ .

## Q3: Best buy

450 g jar:  $285\text{p} \div 450\text{ g} = 0.633\text{p}$  per gram.

720 g jar:  $432\text{p} \div 720\text{ g} = 0.600\text{p}$  per gram.

The 720 g jar is cheaper per gram, so the **720 g jar is better value**.

## Q4: Simultaneous equations

Add the two equations:  $(3x + 2y) + (5x - 2y) = 17 + 7 \Rightarrow 8x = 24 \Rightarrow \mathbf{x = 3}$ .

Substitute into the first equation:  $3(3) + 2y = 17 \Rightarrow 2y = 8 \Rightarrow \mathbf{y = 4}$ .

## Q5: Frustum volume

By similar triangles: if the full cone has base radius 9 cm and the smaller cone has radius 4 cm, then the small cone's height relates to the full cone's height as 4 : 9.

Let full cone height =  $H$ , small cone height =  $h$ . Then  $H - h = 15$  (the frustum height) and  $h / H = 4 / 9$ . So  $h = (4/9)H$ , giving  $H - (4/9)H = 15 \Rightarrow (5/9)H = 15 \Rightarrow H = 27$  cm. Small cone height  $h = 12$  cm.

Volume of full cone =  $(1/3)\pi(9)^2(27) = 729\pi\text{ cm}^3$ .

Volume of small cone =  $(1/3)\pi(4)^2(12) = 64\pi\text{ cm}^3$ .

Frustum volume =  $729\pi - 64\pi = 665\pi\text{ cm}^3 = 2089.51\dots \approx \mathbf{2090\text{ cm}^3}$  (3 s.f.).

## Q6: Quadratic formula

$a = 1$ ,  $b = 5$ ,  $c = -2$ . Discriminant =  $5^2 - 4(1)(-2) = 25 + 8 = 33$ .

$x = (-5 \pm \sqrt{33}) / 2 = (-5 \pm 5.7446) / 2$ .

$x = 0.3723$  or  $x = -5.3723$ .

$\mathbf{x \approx 0.372}$  or  $\mathbf{x \approx -5.37}$  (3 s.f.).

## Q7: Probability with Venn

(a) Students who study French or German =  $32 + 24 - 10 = 46$  (inclusion-exclusion).

Students who study neither =  $60 - 46 = \mathbf{14}$ .

(b) Of the 32 who study French, 10 also study German.

$P(\text{German} \mid \text{French}) = 10 / 32 = \mathbf{5/16}$ .

## Q8: Frequency polygon

(a) Highest frequency is 22, at midpoint 50 g, which is the class  $40 < m \leq 60$ .

(b) Estimate mean using midpoints  $\times$  frequencies:

$$(10 \times 4) + (30 \times 12) + (50 \times 22) + (70 \times 16) + (90 \times 6)$$

$$= 40 + 360 + 1100 + 1120 + 540 = 3160.$$

$$\Sigma f = 4 + 12 + 22 + 16 + 6 = 60.$$

$$\text{Mean} \approx 3160 / 60 = \mathbf{52.7 \text{ g (3 s.f.)}}.$$

### Q9: Indices

(a)  $(3a^2b^4)^3 = 3^3 \times a^6 \times b^{12} = \mathbf{27a^6b^{12}}$ .

(b)  $(16x^8)^{3/4} = 16^{3/4} \times x^6$ . Since  $16^{1/4} = 2$ , we get  $16^{3/4} = 2^3 = 8$ .

$$\text{Answer} = \mathbf{8x^6}.$$

### Q10: Rationalising

Multiply top and bottom by the conjugate  $(3 + \sqrt{3})$ :

$$6(3 + \sqrt{3}) / ((3 - \sqrt{3})(3 + \sqrt{3})) = (18 + 6\sqrt{3}) / (9 - 3) = (18 + 6\sqrt{3}) / 6.$$

$$= \mathbf{3 + \sqrt{3}}. \text{ So } a = 3, b = 1.$$

### Q11: Vectors

(a)  $\mathbf{OM} = (1/2)\mathbf{a}$ .

$$\mathbf{AB} = \mathbf{OB} - \mathbf{OA} = \mathbf{b} - \mathbf{a}.$$

$$\mathbf{AN} = (1/3)\mathbf{AB} = (1/3)(\mathbf{b} - \mathbf{a}).$$

$$\mathbf{ON} = \mathbf{OA} + \mathbf{AN} = \mathbf{a} + (1/3)(\mathbf{b} - \mathbf{a}) = (2/3)\mathbf{a} + (1/3)\mathbf{b}.$$

$$\mathbf{MN} = \mathbf{ON} - \mathbf{OM} = (2/3)\mathbf{a} + (1/3)\mathbf{b} - (1/2)\mathbf{a} = \mathbf{(1/6)a + (1/3)b}.$$

(b) Rewrite:  $(1/6)\mathbf{a} + (1/3)\mathbf{b} = (1/6)(\mathbf{a} + 2\mathbf{b})$ .

$$\text{So } p = 1, q = 2.$$

### Q12: Reverse compound interest

$$\text{Final amount} = \text{Original} \times (1.035)^4 = 1146.25.$$

$$(1.035)^4 = 1.14752... \text{ So Original} = 1146.25 / 1.14752 \approx \mathbf{\pounds 999.00} \approx \pounds 1000 \text{ (to nearest pound).}$$

(Accept  $\pounds 999$  or  $\pounds 1000$ .)

### Q13: Exponential decay

(a)  $M = 80 \times (0.92)^5 = 80 \times 0.6591 = \mathbf{52.7 \text{ g (1 d.p.)}}$ .

(b) Multiplier 0.92 means mass becomes 92% each year, so it decreases by **8%**.

### Q14: Cosine rule

$$BC^2 = AB^2 + AC^2 - 2 \times AB \times AC \times \cos(A).$$

$$= 8^2 + 11^2 - 2 \times 8 \times 11 \times \cos 53^\circ.$$

$$= 64 + 121 - 176 \times 0.6018 = 185 - 105.92 = 79.08.$$

$$BC = \sqrt{79.08} = \mathbf{8.89 \text{ cm (3 s.f.)}}.$$

### Q15: Algebraic fractions

Common denominator =  $(x - 2)(x + 1)$ .

$(2x + 1)(x + 1) - (x - 3)(x - 2)$ , all over  $(x - 2)(x + 1)$ .

Numerator:  $(2x^2 + 3x + 1) - (x^2 - 5x + 6) = x^2 + 8x - 5$ .

Answer:  $(x^2 + 8x - 5) / ((x - 2)(x + 1))$ .

### Q16: Recurring decimal

Let  $x = 0.454545\dots$

Then  $100x = 45.4545\dots$

Subtract:  $100x - x = 45.4545\dots - 0.4545\dots = 45$ .

So  $99x = 45$ , giving  $x = 45/99 = \mathbf{5/11}$ . ✓

### Q17: Cyclic quadrilateral

Opposite angles of a cyclic quadrilateral sum to  $180^\circ$ .

(a) Angle QRS =  $180 - \text{angle SPQ} = 180 - 112 = \mathbf{68^\circ}$ .

(b) Angle RSP =  $180 - \text{angle PQR} = 180 - 84 = \mathbf{96^\circ}$ .

### Q18: Direct proportion

(a)  $y = k\sqrt{x}$ . When  $x = 36$ ,  $y = 24$ :  $24 = k \times 6$ , so  $k = 4$ .

$y = 4\sqrt{x}$ .

(b)  $y = 4\sqrt{81} = 4 \times 9 = \mathbf{36}$ .

### Q19: Box plots

(a) Class A median (the line in the middle of the box) = **52**.

(b) Class B:  $Q1 = 32$ ,  $Q3 = 55$ .  $IQR = 55 - 32 = \mathbf{23}$ .

(c) Comparisons (any two):

- Class A has a higher median (52 vs 45), so on average Class A scored higher.
- Class A has a larger IQR (27 vs 23), so Class A's scores are more spread out (less consistent).
- Class A has a higher maximum (78 vs 70).

### Q20: Quadratic inequality

Factorise:  $x^2 - 5x - 14 = (x - 7)(x + 2)$ .

Critical values:  $x = 7$  and  $x = -2$ .

The parabola opens upward (coefficient of  $x^2$  is positive), so  $(x - 7)(x + 2) > 0$  **outside** the roots.

Solution:  $x < -2$  or  $x > 7$ .

### Q21: Trapezium

(a) Area =  $(1/2)(a + b)h = (1/2)(6 + 14)(5) = \mathbf{50 \text{ cm}^2}$ .

(b) Drop a perpendicular from B to DC, meeting DC at point E. Then  $BE = 5 \text{ cm}$  (the height).

Horizontal distance from B to C: since  $AB = 6$  and  $DC = 14$ , and AD is perpendicular,  $EC = 14 - 6 = 8 \text{ cm}$ .



Apply Pythagoras:  $BC^2 = BE^2 + EC^2 = 5^2 + 8^2 = 25 + 64 = 89$ .

$BC = \sqrt{89} = \mathbf{9.43 \text{ cm (3 s.f.)}}$ .

### Q22: Algebraic proof

Let two consecutive even integers be  $2n$  and  $2n + 2$ , where  $n$  is any integer.

Product =  $2n \times (2n + 2) = 4n^2 + 4n = 4(n^2 + n) = 4n(n + 1)$ .

Since  $n$  is an integer,  $n(n + 1)$  is also an integer.

Therefore  $4n(n + 1)$  is always a **multiple of 4**. ✓

*(In fact this is always a multiple of 8, since  $n(n + 1)$  is always even — consecutive integers.)*